Weak periodic phase shaping lowers the worst-case sensitivity of coherent-state qubit readout

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A fixed interaction can encode the real amplitude of a coherent bosonic mode into a qubit, with the field subsequently discarded. The optimal linear quadrature coupling gives amplitude-independent quantum Fisher information 4/e. We show that every fixed real bounded periodic phase perturbation along the same qubit rotation axis, nonconstant almost everywhere, lowers the infimum of the symmetric-logarithmic-derivative quantum Fisher information over all real amplitudes, for both sufficiently small nonzero signs of its strength. A Gaussian-score calculation gives a nonzero, zero-mean first-order sensitivity response whose Fourier multipliers have no nonzero-frequency zeros. Explicit uniform rank and quadratic-remainder bounds convert this response into a strict worst-case loss. The argument includes discontinuous profiles and requires neither a photon cutoff nor a sampled-amplitude extrapolation. The strength radius depends on the fixed profile and period. This restricted obstruction excludes a weak periodic shaping route; it does not determine the unrestricted interaction optimum or exclude finite-strength improvements.

I. INTRODUCTION

When a quantized field is measured through a small probe, the interaction must transfer information into the degrees of freedom that are actually read out. For the real amplitude α of a single coherent mode, the field quantum Fisher information (QFI) is 4, whereas a fixed qubit readout generally retains less. Ravell Rodríguez, Perarnau-Llobet, and Sekatski formulated the optimization of the worst-amplitude qubit QFI and exhibited a linear quadrature coupling with uniform value 4/e [1]. Their work leaves the general fixed-interaction optimization unresolved and suggests 4/e or a nearby value.

A natural design question is whether weak nonlinear shaping of that linear interaction can improve its uniform sensitivity. Optimizing the linear coupling strength alone does not answer this question: nonlinear profiles introduce infinitely many additional directions. Here we analyze each fixed bounded periodic profile acting along the same qubit rotation axis. Every nonconstant profile produces a first-order ripple in the local QFI, and that ripple necessarily has both signs. A uniform remainder estimate then proves a strict reduction of the worst-amplitude QFI for either sufficiently small sign of the perturbation.

This is a directionwise statement. The profile and its period are fixed before the perturbation strength tends to zero. The result does not provide a common neighborhood over all profiles, an upper bound for finite-strength shaping, or a solution of the arbitrary-unitary problem.

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II. MODEL AND THE LINEAR BENCHMARK

The field Hilbert space is $\ell^2(\mathbb{N}_0)$, with $[a, a^\dagger] = 1$. In oscillator units, $\hbar = 1$ and

$$Q = \frac{a + a^\dagger}{\sqrt{2}}, \qquad p_\alpha(x) = \frac{\mathrm{e}^{-(x-\sqrt{2}\alpha)^2}}{\sqrt{\pi}}. \tag{1}$$

The input is the real coherent state $|\alpha\rangle$, so its quadrature has mean $\sqrt{2}\alpha$ and variance 1/2. The amplitude, quadrature, and couplings below are dimensionless. The qubit begins in the fixed state $|g\rangle$. In the basis $(|g\rangle, |e\rangle)$, use $\sigma_y = \left(\begin{smallmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{smallmatrix}\right)$, so $\mathrm{e}^{-\mathrm{i}\sigma_y f(x)}|g\rangle = \cos f(x)|g\rangle + \sin f(x)|e\rangle$.

Let $L > 0$ and let g be a real bounded Borel L-periodic function, fixed independently of α and of the strength ϵ. We consider

$$V_\epsilon = \exp\left[-\mathrm{i}\sigma_y \otimes \left(\frac{Q}{\sqrt{2}} + \epsilon g(Q)\right)\right]. \tag{2}$$

Since $g(Q)$ is bounded and self-adjoint, the quadrature function in Eq. (2) is self-adjoint on the domain of Q. Functional calculus defines a unitary on the full field space, including for discontinuous g. Only the final qubit is retained:

$$\begin{aligned} \rho_\epsilon(\alpha) &= \mathrm{Tr}_F\left[V_\epsilon(|g\rangle\langle g| \otimes |\alpha\rangle\langle\alpha|)V_\epsilon^\dagger\right] \\ &= \int_{\mathbb{R}} p_\alpha(x)|\psi_\epsilon(x)\rangle\langle\psi_\epsilon(x)|\,\mathrm{d}x, \\ |\psi_\epsilon(x)\rangle &= \mathrm{e}^{-\mathrm{i}\sigma_y[x/\sqrt{2}+\epsilon g(x)]}|g\rangle. \end{aligned} \tag{3}$$

There is one input mode per encoding, no postselection or retained classical branch record, and no control adapted to the unknown amplitude.

We use the pointwise symmetric logarithmic derivative (SLD) QFI [2], defined by

$$\rho_\epsilon\mathcal{L}_\alpha + \mathcal{L}_\alpha\rho_\epsilon = 2\partial_\alpha\rho_\epsilon, \qquad F_Q = \mathrm{Tr}(\rho_\epsilon\mathcal{L}_\alpha^2). \tag{4}$$

This is a local sensitivity benchmark, allowing optimization of the final qubit measurement at a local working point. It does not imply a globally identifiable amplitude or a particular finite-sample estimation risk. Uniform full rank, established below, avoids rank-changing SLD conventions.

Put $\mu = \sqrt{2}\alpha$ and $p_0(y) = \mathrm{e}^{-y^2}/\sqrt{\pi}$. The complex Bloch coordinate is

$$m_\epsilon(\mu) = \int_{\mathbb{R}} p_0(y)\mathrm{e}^{\mathrm{i}\sqrt{2}(\mu+y)+2\mathrm{i}\epsilon g(\mu+y)}\,\mathrm{d}y. \tag{5}$$

In particular,

$$\rho_\epsilon = \frac{1}{2}\begin{pmatrix} 1+\mathrm{Re}\, m_\epsilon & \mathrm{Im}\, m_\epsilon \\ \mathrm{Im}\, m_\epsilon & 1-\mathrm{Re}\, m_\epsilon \end{pmatrix}. \tag{6}$$

Define $d_\epsilon(\mu) = \sqrt{2}\partial_\mu m_\epsilon(\mu)$ and $\mathcal{F}_\epsilon(\mu) = F_Q[\rho_\epsilon(\mu/\sqrt{2})]$. Thus d_ϵ is always an *amplitude* derivative. Solving Eq. (4) in the Bloch representation gives

$$\mathcal{F}_\epsilon(\mu) = |d_\epsilon|^2 + \frac{[\mathrm{Re}(\overline{m}_\epsilon d_\epsilon)]^2}{1-|m_\epsilon|^2}. \tag{7}$$

For completeness, writing $\mathcal{L}_\alpha = b_0 I + \boldsymbol{b}\cdot\boldsymbol{\sigma}$ gives $b_0 + \boldsymbol{b}\cdot\boldsymbol{r} = 0$ and $\boldsymbol{b} + b_0\boldsymbol{r} = \partial_\alpha \boldsymbol{r}$, which immediately yields Eq. (7) for $\boldsymbol{r} = (\mathrm{Im}\, m, 0, \mathrm{Re}\, m)$.

More generally, the linear coupling $f(x) = \tau x/\sqrt{2}$ gives

$$m_\tau(\alpha) = \mathrm{e}^{-\tau^2/2+2\mathrm{i}\tau\alpha}, \qquad F_Q = 4\tau^2\mathrm{e}^{-\tau^2}. \tag{8}$$

This is the exact quadrature benchmark of Ref. [1], not a large-amplitude approximation to the Jaynes–Cummings interaction. It is maximized at $|\tau| = 1$, giving $4/\mathrm{e}$ for every real α. Equation (2) perturbs the $\tau = 1$ carrier.

III. THE FIRST-ORDER SENSITIVITY RESPONSE

The derivative in Eq. (7) can be obtained by differentiating the Gaussian density at fixed quadrature x:

$$d_\epsilon(\mu) = \int_{\mathbb{R}} 2\sqrt{2}\, y\, p_0(y)\mathrm{e}^{\mathrm{i}\sqrt{2}(\mu+y)+2\mathrm{i}\epsilon g(\mu+y)}\,\mathrm{d}y. \tag{9}$$

The integrable Gaussian score justifies this differentiation even when g has jumps; no derivative of g is used.

Let $A = \mathrm{e}^{-1/2}$. At zero strength,

$$m_0 = A\mathrm{e}^{\mathrm{i}\sqrt{2}\mu}, \qquad d_0 = 2\mathrm{i}A\mathrm{e}^{\mathrm{i}\sqrt{2}\mu}. \tag{10}$$

Hence $|d_0|^2 = 4/\mathrm{e}$ and $\mathrm{Re}(\overline{m}_0 d_0) = 0$. If $m_1 = \partial_\epsilon m_\epsilon|_0$ and $d_1 = \partial_\epsilon d_\epsilon|_0$, then

$$\begin{aligned} m_1 &= 2\mathrm{i}\int p_0(y)g(\mu+y)\mathrm{e}^{\mathrm{i}\sqrt{2}(\mu+y)}\,\mathrm{d}y, \\ d_1 &= 4\mathrm{i}\sqrt{2}\int y p_0(y)g(\mu+y)\mathrm{e}^{\mathrm{i}\sqrt{2}(\mu+y)}\,\mathrm{d}y. \end{aligned} \tag{11}$$

The radial term in Eq. (7) starts at second order. Therefore the first variation is

$$\begin{aligned} G(\mu) &= \partial_\epsilon \mathcal{F}_\epsilon(\mu)|_0 = 2\,\mathrm{Re}(\overline{d}_0 d_1) \\ &= 16\sqrt{2}A\int_{\mathbb{R}} y p_0(y)\cos(\sqrt{2}y)g(\mu+y)\,\mathrm{d}y. \end{aligned} \tag{12}$$

The kernel is smooth, integrable, and odd. Its convolution with bounded g is continuous, L-periodic, and has zero mean.

To determine whether any nonconstant profile can have zero response, write $\kappa = 2\pi/L$ and

$$c_n = \frac{1}{L}\int_0^L g(x)\mathrm{e}^{-\mathrm{i}n\kappa x}\,\mathrm{d}x. \tag{13}$$

The Gaussian moment $\int y p_0(y)\mathrm{e}^{\mathrm{i}\lambda y}\,\mathrm{d}y = \mathrm{i}\lambda\mathrm{e}^{-\lambda^2/4}/2$ gives, for $n > 0$,

$$G_n = \mathrm{i}C(n\kappa)c_n, \qquad G_{-n} = -\mathrm{i}C(n\kappa)c_{-n}, \qquad G_0 = 0, \tag{14}$$

where $k_0 = 1/\sqrt{2}$ and

$$\begin{aligned} C(\xi) &= 16k_0\mathrm{e}^{-1-\xi^2/4} \\ &\quad\times\left[\xi\cosh(k_0\xi) - 2k_0\sinh(k_0\xi)\right]. \end{aligned} \tag{15}$$

Fubini's theorem applies because g is bounded and the kernel is integrable. No pointwise Fourier-series convergence of g is assumed.

For $\xi > 0$, put $z = k_0\xi$. The bracket in Eq. (15) is $\sqrt{2}(z\cosh z - \sinh z)$, which is strictly positive: it vanishes at $z = 0$ and has derivative $\sqrt{2}z\sinh z > 0$. Thus $C(\xi) > 0$ at every nonzero positive frequency. If G vanished identically, Eq. (14) would force all nonzero Fourier coefficients of g to vanish. Fourier uniqueness in $L^2([0, L])$ would then make g constant almost everywhere. For a nonconstant profile, it follows that

$$\begin{aligned} \min_\mu G(\mu) < 0 < \max_\mu G(\mu), \\ \gamma = \min\{-\min G, \max G\} > 0. \end{aligned} \tag{16}$$

The extrema exist on any compact period. Each fixed perturbation therefore has an amplitude at which its first-order QFI change is unfavorable, for either sign of strength.

For a finite trigonometric profile,

$$g(x) = g_0 + \sum_{n\geq 1}[a_n\cos(n\kappa x) + b_n\sin(n\kappa x)], \tag{17}$$

the response is explicitly

$$G(\mu) = \sum_{n\geq 1} C(n\kappa)[b_n\cos(n\kappa\mu) - a_n\sin(n\kappa\mu)]. \tag{18}$$

Distinct harmonics cannot cancel the entire ripple: no nonzero mode lies in the response kernel.

IV. UNIFORM RANK CONTROL AND STRICT WORST-CASE LOSS

A first variation at isolated amplitudes would not by itself control the whole real axis. We next bound the QFI remainder uniformly. Choose any finite $M > 0$ satisfying $|g(x)| \le M$, and define

$$D = 2A, \qquad W = 2\sqrt{2/\pi}, \qquad h = (1-A)/4,$$
$$\beta = (1+A)/2, \qquad \delta = 1-\beta^2,$$
$$K = 4W^2(1+h)^2 + 4DW + \frac{4(\beta W + D)^2}{\delta}. \tag{19}$$

All these constants are positive, and $\beta < 1$.

Write $a = |\epsilon|$. The inequalities $|e^{2i\epsilon g} - 1| \le 2aM$ and $\int 2\sqrt{2}|y|p_0(y)\,dy = W$ imply

$$|m_\epsilon - m_0| \le 2aM, \qquad |d_\epsilon - d_0| \le 2aMW. \tag{20}$$

When $aM \le h$, these give

$$|m_\epsilon| \le A + 2h = \beta, \qquad 1 - |m_\epsilon|^2 \ge \delta. \tag{21}$$

The smallest output eigenvalue is at least $(1-\beta)/2 = h$, independently of the amplitude.

For real g, the exponential Taylor remainder obeys $|e^{2i\epsilon g} - 1 - 2i\epsilon g| \le 2a^2M^2$. Consequently,

$$m_\epsilon = m_0 + \epsilon m_1 + r_m, \qquad |r_m| \le 2a^2M^2,$$
$$d_\epsilon = d_0 + \epsilon d_1 + r_d, \qquad |r_d| \le 2a^2M^2W,$$
$$|m_1| \le 2M, \qquad |d_1| \le 2MW. \tag{22}$$

Using $|d_0| = D$ and $G = 2\,\mathrm{Re}(\overline{d}_0 d_1)$, expansion of the first term of Eq. (7) yields

$$\big||d_\epsilon|^2 - D^2 - \epsilon G\big| \le 2D|r_d| + |\epsilon d_1 + r_d|^2$$
$$\le a^2M^2[4DW + 4W^2(1+h)^2]. \tag{23}$$

For the second term, the zero radial derivative at the carrier and Eq. (20) give

$$|\,\mathrm{Re}(\overline{m}_\epsilon d_\epsilon)| \le |m_\epsilon|\,|d_\epsilon - d_0| + |m_\epsilon - m_0|\,|d_0|$$
$$\le 2aM(\beta W + D). \tag{24}$$

Combining Eqs. (21), (23), and (24) proves the exact bound

$$\left|\mathcal{F}_\epsilon(\mu) - \frac{4}{e} - \epsilon G(\mu)\right| \le KM^2\epsilon^2 \quad \text{for all } \mu \in \mathbb{R}, \tag{25}$$

whenever $|\epsilon|M \le h$.

Proposition 1 *For each fixed $L > 0$ and real bounded Borel L-periodic profile g that is not constant almost everywhere, let γ and K be given by Eqs. (16) and (19). With*

$$\epsilon_0(g, L) = \min\left\{\frac{h}{M}, \frac{\gamma}{2KM^2}\right\} > 0, \tag{26}$$

every $0 < |\epsilon| < \epsilon_0$, of either sign, gives uniformly full-rank qubit outputs and

$$\inf_{\alpha\in\mathbb{R}} F_Q[\rho_\epsilon(\alpha)] < \frac{4}{e}. \tag{27}$$

Proof. For $\epsilon > 0$, evaluate Eq. (25) at a minimum of G. For $\epsilon < 0$, use a maximum of G. In either case, using $\mu = \sqrt{2}\alpha$ and Eq. (26),

$$\inf_{\alpha\in\mathbb{R}} F_Q[\rho_\epsilon(\alpha)] \le \frac{4}{e} - \gamma|\epsilon| + KM^2\epsilon^2$$
$$< \frac{4}{e} - \frac{\gamma}{2}|\epsilon| < \frac{4}{e}. \tag{28}$$

Equation (21) supplies the uniform rank claim. □

If g is constant almost everywhere, the perturbation is a fixed output rotation and preserves 4/e exactly. Thus the exclusion of constant profiles is necessary, and changes on sets of measure zero are immaterial.

V. ILLUSTRATION AND NUMERICAL CONSISTENCY CHECKS

For $g(x) = \sin(\kappa x)$, Eq. (18) reduces to $G(\mu) = C(\kappa)\cos(\kappa\mu)$. Figure 1 illustrates the response and the opposite unfavorable amplitude regions for the two signs. Its QFI values were calculated from the Gaussian projector and score integrals, followed by a spectral SLD evaluation. They are supporting numerical data; the proposition follows from the uniform analytic bound. The illustrated strengths satisfy the rank/remainder condition $|\epsilon|M \le h$; they need not lie within the conservative strict-loss radius in Eq. (26).

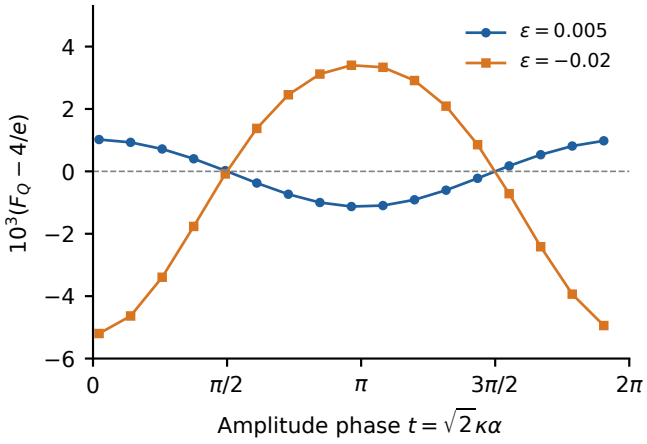


FIG. 1. Sensitivity ripple for the fixed profile $g(x) = \sin(\kappa x)$ with $\kappa = 0.7$, plotted against $t = \kappa\mu = \sqrt{2}\kappa\alpha$. QFI change from 4/e at $\epsilon = 0.005$ and $\epsilon = -0.02$, from direct density-matrix SLD calculations at 17 phases. Lines join sampled values to guide the eye. Both curves have values below the baseline. No grid minimum or numerical error enclosure is used to prove the all-amplitude statement.

The same consistency tests used a mixed-harmonic profile, a discontinuous square wave, a high harmonic, and a constant control. Across five profiles and 17 phases at two strengths, the 170 direct SLD evaluations satisfy the analytic rank and remainder inequalities. The

largest discrepancy between the Gaussian and Fourier first-response calculations is below 3.8×10^{-15}, and the largest observed ratio of the absolute QFI remainder to $KM^2\epsilon^2$ is below 0.057. The square-wave integral is split at its jumps. These finite floating-point checks do not provide outward-rounded quadrature certificates or establish the response's all-frequency injectivity. Equations (15)–(25) provide the corresponding exact statements. Code, the saved data, and regeneration instructions accompany this draft.

VI. PHYSICAL CONSEQUENCE AND LIMITATIONS

The loss mechanism is a nonzero first-order modulation with zero average: weak shaping can increase the QFI at some amplitudes, but decreases it at others. The response filter is injective on nonconstant periodic profiles, so adding harmonics cannot eliminate the entire first-order modulation. This rules out every fixed weak bounded periodic same-axis profile, including profiles with jumps, as a route to improving the established uniform qubit readout.

The result concerns one fixed interaction over the whole amplitude axis. Locally optimized preprocessing under noisy measurements, such as the finite-dimensional setting of Ref. [3], has a different objective and does not furnish this uniform fixed-profile conclusion. Likewise, retaining an additional measurement record or querying several field modes would change the readout resources specified here.

The radius in Eq. (26) depends on g and L. The multipliers can be small at low or high frequencies, so the present argument supplies no common radius over all periods or profiles. Finite-strength deformations, strength-dependent profiles, and interactions outside the same-axis functional form remain beyond the proposition. Their possible success is not established. The unrestricted fixed-unitary worst-amplitude optimization is not resolved by this result, and no experimental implementation or global estimation advantage is claimed.

USE OF AI TOOLS

OpenAI Codex was used substantively for literature retrieval, derivation development, symbolic and numerical calculations, code preparation, manuscript drafting, and internal checking. Separate AI sessions reconstructed the source model and recomputed the restricted result. These checks are not external peer review or human scientific verification. Human scientific review and authorship confirmation remain pending for this local draft; the accompanying author handoff records the disclosure details that require confirmation.

DATA AND CODE AVAILABILITY

The accompanying reproducibility material contains the analytic argument, numerical consistency-test code and saved outputs, figure-generation script, software versions, and build instructions. No experimental data were used.

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